

H13 1.2344

Name:

X 40 CrMoV 5-1

Material No.:

1.2344

ISI H13

Typical analysis in %:

C	Si	Cr	Mo	V
0.4	1.1	5.2	1.4	0.95

As-supplied condition:

Soft-annealed to max. 235 HB
(791 N/mm²)

Characteristics:

CrMoV alloyed hot work steel with excellent high temperature strength and good toughness, good high temperature wear resistance, highest thermal shock resistance, very good cleanliness factor

and excellent homogeneity, nitridable

General fields of application:

Tools for forging machines, dies, die inserts, extrusion tools, hot shearing knives and tools for plastics processing

Special note:

If nitrided, the nitriding depth should not be too deep otherwise increased thermal cracking may occur.

Preheating to 200 - 300 °C before starting work is recommended.

Also available in ESR grade .

We recommend H13 1.2344 steel in ESR grade for highly polished mirrored finishes.

Heat treatment data:

	Temperature	Duration	Cooling
Soft annealing	800 - 860 °C	2 - 5 h	furnace
Stress-relief annealing	600 - 650 °C	min. 4 h	furnace
Hardening	1020 - 1060 °C	Group II	oil, air WB 500 °C
Tempering	530 - 700 °C 3 x, see tempering curve	min. 2 h depending on cross section	still air

Physical characteristics:

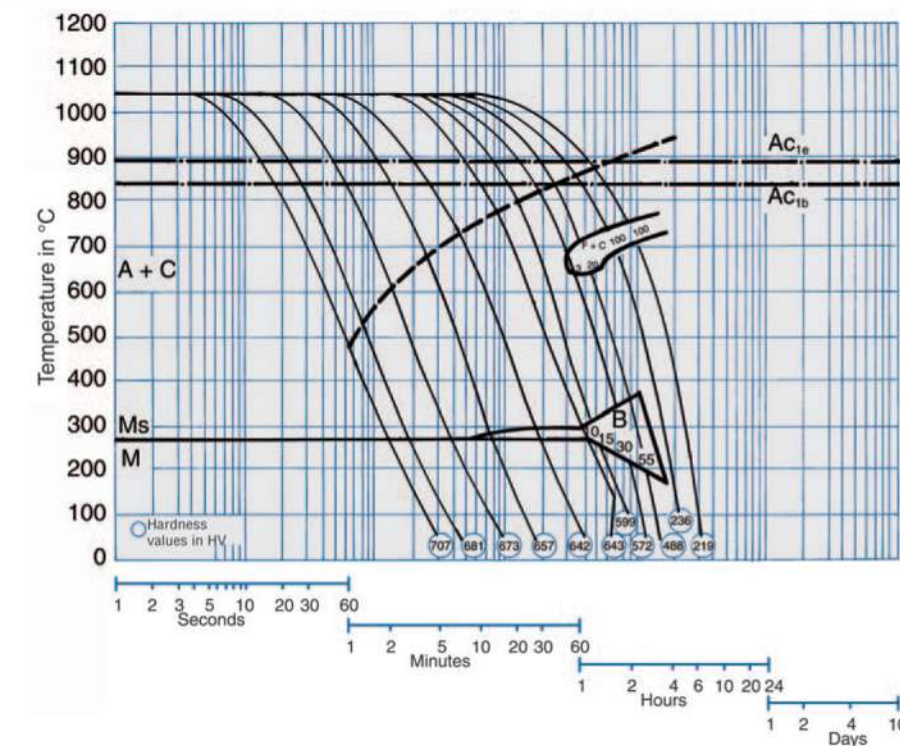
Coefficient of thermal expansion: between 20 °C and:

10 ⁻⁶ x m	100	200	300	400	500	600	700 °C
m x K	10.9	11.9	12.2	12.5	12.9	13	13.5

Thermal conductivity:	W	20	350	600 °C
m x K		25	26.8	29.3

Normal working hardness: 30 - 54 HRC (1000 - 1900 N/mm²)

Continuous time-temperature-transformation diagram



Tempering curve

